Žiga Kovačič

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Scholar: Google Scholar

EDUCATION

Sep. 2026 – Present — **Stanford University**
Ph.D. in Computer Science
Advisor: Jiajun Wu

Aug. 2022 – May 2026 — **Cornell University**
B.A. in Computer Science (Honors) and College Scholars (World Modeling)
Advisor: Kevin Ellis
Final GPA: 4.12/4.0

PUBLICATIONS

2026 — **MPMWorlds: Material-Point-Method Simulations for Inferring and Extrapolating Physical Dynamics**
Ž. Kovačič, K. Ellis.

Book 2026 — **Physics-Based Simulation**
M. Li, C. Jiang, Z. Luo, **Ž. Kovačič**, W. Du, C. Yu, T. Xie.

NeurIPS 2025 — **Refactoring Codebases through Library Design**
Ž. Kovačič*, C. Lee*, J. Chiu, W. Zhao, K. Ellis.

SIGGRAPH 2025 — **Pocket Time-Lapse**
E. Chen, **Ž. Kovačič**, M. Aggarwal, A. Davis.

OTHER RESEARCH

Summer 2025 – Present — **Cornell Material Simulation Group**
Advisor: Prof. Steve Marschner

Multi-scaled Yarn Simulation
Implemented an MPM-based yarn model to study multi-scale fiber interactions and evaluated reduced-order methods (e.g., CROM) for accelerating simulation. Explored how hybrid data-driven simulators can generalize across yarn geometries and fiber configurations.

Spring 2024 – Jan 2025 — **Cornell Graphics and HCI Lab**
Advisor: Prof. Abe Davis

Interactive Image-Space Modal Re-simulation
Reimplemented ISMB for real-time video-based re-simulation using image-space modal analysis; extended the method with modal warping to address artifacts of linear modal analysis.

TEACHING EXPERIENCE

Spring 2026 — **Teaching Assistant (Head TA)**, Cornell University
CS 4782: Introduction to Deep Learning

Fall 2025 — **Teaching Assistant (Head TA)**, Cornell University
CS 4787: Large Scale Machine Learning

Spring 2025	**Teaching Assistant (Head TA)**, Cornell University CS 4782: Introduction to Deep Learning. Built the autograder for the class.
Fall 2024	**Teaching Assistant**, Cornell University CS 4620: Introduction to Computer Graphics
Spring 2024	**Teaching Assistant**, Cornell University CS 4780: Introduction to Machine Learning

LEADERSHIP AND MENTORSHIP

Fall 2024 – Spring 2025	**Vice-President**, Cornell University Artificial Intelligence (CUAI) Responsible for leading and mentoring a team of 17 undergraduate researchers. Fostering an environment for undergrad-led research. Organizing a weekly reading group on recent papers for undergrads.

HONORS

May 2026	**Merrill Scholar**, Cornell University Top 1% of graduating class
2025 – 2026	**Honorable mention**, CRA Outstanding Undergraduate Researcher Award
Spring 2024	**TA Award**, Cornell Computer Science Department
Fall 2022 – Spring 2025	**Dean's List**, Cornell University

PROJECTS

Oct 2024 – Feb 2025	**Pen-2-Graph** Developed a novel pipeline using Vision-Language Models, Differential Evolution, and mathematical constraints on graph structure to automatically synthesize node-edge graph programs from hand-drawn sketches.
Mar 2025 – May 2025	**Material Point Method and CPIC** Engineered a real-time 2D/3D MLS-MPM solver with CPIC method for rigid-deformable object interactions using Taichi, simulating elastic, fluid, plastic, sand, and visco-plastic materials on GPU-backed kernels. Connected simulation output to Blender for lighting/rendering workflows, enabling end-to-end high-quality renders from physics simulation pipelines via Taichi-Blender integration.
Oct 2024 – Feb 2025	**SliceSplatting – Obstruction Removal from 3D reconstructions** Modified Gaussian Splatting to automatically remove obstructions (e.g. fences) in 3D reconstructions using depth-aware particle slicing. Able to remove multi-layers of obstructions, improving upon previous NeRF-based methods.

SKILLS

Languages:	Python, JavaScript, TypeScript, C/C++, HTML/CSS
Libraries:	PyTorch, Taichi, Numpy, open-cv, WebGL, THREE.js

RELEVANT COURSES

(Grad) Physically Based Animations (A+), (Grad) Program Synthesis (A+), (Grad) 3D Computer Vision (A+), Reinforcement Learning (A), (Grad) Computation for Content Creation (A), (Grad) Computational Imaging (A), Graphics (A+), Machine Learning (A+), Algorithms (A+), Honors Real Analysis II (A+), Numerical Analysis (A)